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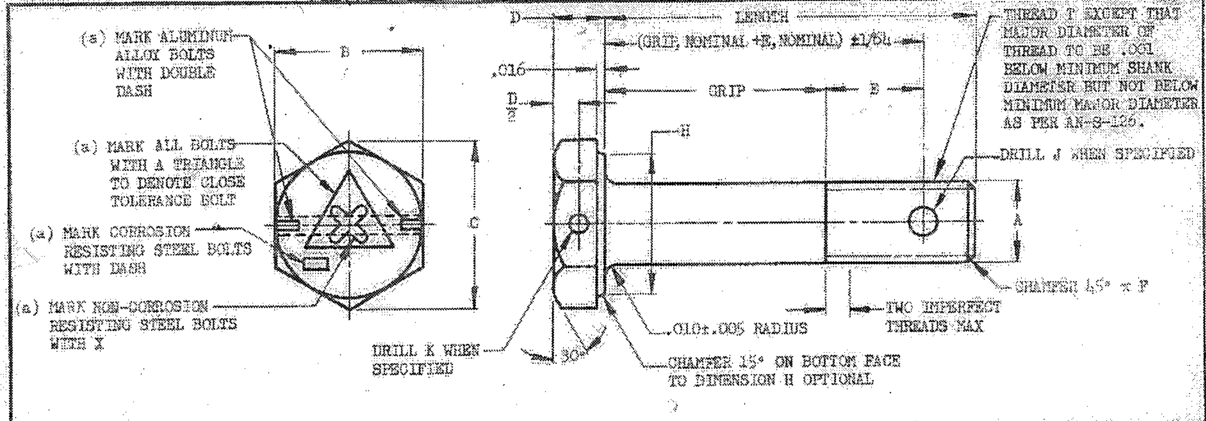
(a) MARK ALUMINUM ALLOY BOLTS WITH DOUBLE DASH
(a) MARK ALL BOLTS WITH A TRIANGLE TO DENOTE CLOSE TOLERANCE BOLT
(a) MARK CORROSION RESISTING STEEL BOLTS WITH DASH
(a) MARK NON-CORROSION RESISTING STEEL BOLTS WITH X
B
C
D
LENGTH
(GRIP, NOMINAL +E, NOMINAL) ±1/64
.016
D/2
GRIP
E
H
A
THREAD T EXCEPT THAT MAJOR DIAMETER OF THREAD TO BE .001 BELOW MINIMUM SHANK DIAMETER BUT NOT BELOW MINIMUM MAJOR DIAMETER AS PER AN-S-126.
DRILL J WHEN SPECIFIED
CHAMFER 45° x F
TWO IMPERFECT THREADS MAX
.010±.005 RADIUS
CHAMFER 15° ON BOTTOM FACE TO DIMENSION H OPTIONAL
DRILL K WHEN SPECIFIED
30°

BASIC AN PART NUMBER	THREAD T	A DIA	B	C REF	D ±.016	E REF	F ±.016	H ±.016 DIA	J	K	RATED STRENGTH (POUNDS) ULTIMATE TENSILE AT ROOT DIA STEEL	ULTIMATE TENSILE AT ROOT DIA AL ALLOY	YIELD TENSILE AT ROOT DIA STEEL	YIELD TENSILE AT ROOT DIA AL ALLOY	SINGLE SHEAR AT FULL DIA STEEL	SINGLE SHEAR AT FULL DIA AL ALLOY
AN173	NO10-32 NF-3	$.1894^{+.0000}_{-.0005}$	$.375^{+.002}_{-.010}$	.430	.125	17/64	.031	.375	.070	.046	2 210	1 100	1 690	710	2 125	990
AN174	1/4 -28 NF-3	$.2492^{+.0000}_{-.0005}$	$.438^{+.002}_{-.010}$	.510	.156	5/16	.031	.438	.076	.046	4 080	2 030	3 130	1 31[illegible]	3 680	1 715
AN175	5/16-24 NF-3	$.3117^{+.0000}_{-.0005}$	$.500^{+.002}_{-.010}$	.580	.188	23/64	.047	.500	.076	.070	6 500	3 220	4 950	2 080	5 750	2 685
AN176	3/8 -24 NF-3	$.3742^{+.0000}_{-.0005}$	$.563^{+.002}_{-.010}$	.650	.219	7/16	.047	.563	.106	.070	10 100	5 020	7 740	3 240	8 280	3 870
AN177	7/16-20 NF-3	$.4367^{+.0000}_{-.0005}$	$.625^{+.002}_{-.010}$	.720	.250	31/64	.047	.625	.106	.070	13 600	6 750	10 430	4 350	11 250	5 250
AN178	1/2 -20 NF-3	$.4991^{+.0000}_{-.0005}$	$.750^{+.002}_{-.010}$	.870	.281	39/64	.047	.750	.106	.070	18 500	9 180	14 190	5 920	14 700	6 850
AN179	9/16-18 NF-3	$.5616^{+.0000}_{-.0005}$	$.875^{+.002}_{-.010}$	1.010	.312	21/32	.062	.875	.141	.070	23 600	11 700	18 100	7 550	18 700	8 700
AN180	5/8 -18 NF-3	$.6240^{+.0000}_{-.0006}$	$.938^{+.002}_{-.010}$	1.090	.344	47/64	.062	.938	.141	.070	30 100	14 900	23 080	9 610	23 000	10 750
AN182	3/4- 16 NF-3	$.7489^{+.0000}_{-.0007}$	$1.063^{+.003}_{-.010}$	1.230	.406	7/8	.062	1.063	.141	.070	44 000	21 800	33 730	14 100	33 150	15 500
AN184	7/8- 14 NF-3	$.8737^{+.0000}_{-.0008}$	$1.250^{+.003}_{-.010}$	1.440	.469	63/64	.078	1.250	.141	.070	60 000	29 800	46 000	19 200	45 050	21 050
AN186	1-14 NF-3	$.9985^{+.0000}_{-.0010}$	$1.438^{+.003}_{-.010}$	1.660	.531	1- 3/32	.078	1.438	.141	.070	80 700	40 000	61 870	25 800	58 900	27 500

(a) MARKING MAY BE EITHER RAISED OR RECESSED BUT SHALL NOT EXCEED .010 INCH IN HEIGHT OR DEPTH.

MATERIAL: NON-CORROSION RESISTING STEEL, CORROSION RESISTING STEEL OR ALUMINUM ALLOY. SEE PROCUREMENT SPECIFICATION.

FINISH: SEE PROCUREMENT SPECIFICATION, EXCEPT AS NOTED. AFTER GRINDING CADMIUM PLATED BOLTS, SHANKS SHALL BE COATED WITH COMPOUND IN ACCORDANCE WITH SPECIFICATION AN-C-52.

ADD C BEFORE DASH NUMBER FOR CORROSION RESISTING STEEL BOLT.
ADD DD BEFORE DASH NUMBER FOR ALUMINUM ALLOY BOLT.
ADD A AFTER DASH NUMBER FOR UNDRILLED BOLT. SEE ILLUSTRATION.
ADD H BEFORE DASH NUMBER FOR BOLT WITH DRILLED HEAD AND SHANK. SEE ILLUSTRATION.
ADD H BEFORE DASH NUMBER AND A AFTER DASH NUMBER FOR BOLT WITH DRILLED HEAD ONLY. SEE ILLUSTRATION.

EXAMPLES OF PART NUMBERS:

AN176-10 = 3/8 NON-CORROSION RESISTING STEEL BOLT 1-5/64 LONG, 7/16 GRIP WITH DRILLED SHANK ONLY. SEE ILLUSTRATION.
AN176C10 = 3/8 CORROSION RESISTING STEEL BOLT 1-5/64 LONG, 7/16 GRIP WITH DRILLED SHANK ONLY. SEE ILLUSTRATION.
AN176DD10 = 3/8 ALUMINUM ALLOY BOLT 1-5/64 LONG, 7/16 GRIP WITH DRILLED SHANK ONLY. SEE ILLUSTRATION.
AN176DD10A = 3/8 ALUMINUM ALLOY BOLT 1-5/64 LONG, 7/16 GRIP, UNDRILLED SHANK AND HEAD. SEE ILLUSTRATION.
AN176DDH10 = 3/8 ALUMINUM ALLOY BOLT 1-5/64 LONG, 7/16 GRIP WITH DRILLED HEAD AND SHANK. SEE ILLUSTRATION.
AN176DDH10A = 3/8 ALUMINUM ALLOY BOLT 1-5/64 LONG, 7/16 GRIP WITH DRILLED HEAD ONLY. SEE ILLUSTRATION.

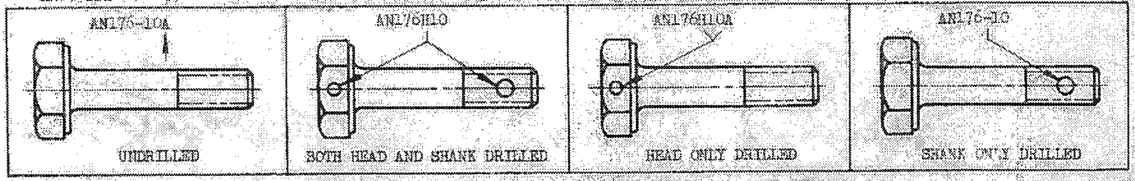
AN176-10A
UNDRILLED
AN176H10
BOTH HEAD AND SHANK DRILLED
AN176H10A
HEAD ONLY DRILLED
AN176-10
SHANK ONLY DRILLED

ILLUSTRATION OF DRILLED AND UNDRILLED BOLTS AND PART NUMBERS

REMOVE BURRS AND SHARP EDGES.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.010; ANGLES ±1°.

DRILLS IN ACCORDANCE WITH AND10387.

APPROVED 29 MAY 48 REVISED

PROCUREMENT SPECIFICATION	AIR FORCE-NAVY AERONAUTICAL STANDARD	AN173 THRU AN186
AN-B-3	BOLT—AIRCRAFT, CLOSE TOLERANCE	SHEET 1 OF 3 SHEETS

SUPERSEDES USAF DRAWINGS
83 THRU 96 AND NAS DRAWINGS
53 THRU 66

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DASH NO.	AN173		AN174		AN175		AN176		AN177		AN178		AN179	
	GRIP ±1/64	LENGTH +1/32 -1/64	GRIP ±1/64	LENGTH +1/32 -1/64	GRIP ±1/64	LENGTH +1/32 -1/64	GRIP ±1/64	LENGTH +1/32 -1/64	GRIP ±1/64	LENGTH +1/32 -1/64	GRIP ±1/64	LENGTH +1/32 -1/64	GRIP ±1/64	LENGTH +1/32 -1/64
3	1/16	15/32	1/16	15/32										
4	1/8	17/32	1/16	17/32	1/16	19/32								
5	1/4	21/32	3/16	21/32	3/16	23/32	1/16	45/64	1/16	23/32				
6	3/8	25/32	5/16	25/32	5/16	27/32	3/16	53/64	3/16	27/32	1/16	27/32	1/16	31/32
7	1/2	29/32	7/16	29/32	7/16	31/32	5/16	61/64	5/16	31/32	3/16	31/32	1/8	1- 1/32
10	5/8	1- 1/32	9/16	1- 1/32	9/16	1- 3/32	7/16	1- 5/64	7/16	1- 3/32	5/16	1- 3/32	1/4	1- 5/32
11	3/4	1- 5/32	11/16	1- 5/32	11/16	1- 7/32	9/16	1-13/64	9/16	1- 7/32	7/16	1- 7/32	3/8	1- 9/32
12	7/8	1- 9/32	13/16	1- 9/32	13/16	1-11/32	11/16	1-21/64	11/16	1-11/32	9/16	1-11/32	1/2	1-13/32
13	1	1-13/32	15/16	1-13/32	15/16	1-15/32	13/16	1-29/64	13/16	1-15/32	11/16	1-15/32	5/8	1-17/32
14	1-1/8	1-17/32	1- 1/16	1-17/32	1- 1/16	1-19/32	15/16	1-37/64	15/16	1-19/32	13/16	1-19/32	3/4	1-21/32
15	1-1/4	1-21/32	1- 3/16	1-21/32	1- 3/16	1-23/32	1- 1/16	1-45/64	1- 1/16	1-23/32	15/16	1-23/32	7/8	1-25/32
16	1-3/8	1-25/32	1- 5/16	1-25/32	1- 5/16	1-27/32	1- 3/16	1-53/64	1- 3/16	1-27/32	1- 1/16	1-27/32	1	1-29/32
17	1-1/2	1-29/32	1- 7/16	1-29/32	1- 7/16	1-31/32	1- 5/16	1-61/64	1- 5/16	1-31/32	1- 3/16	1-31/32	1-1/8	2- 1/32
20	1-5/8	2- 1/32	1- 9/16	2- 1/32	1- 9/16	2- 3/32	1- 7/16	2- 5/64	1- 7/16	2- 3/32	1- 5/16	2- 3/32	1-1/4	2- 5/32
21	1-3/4	2- 5/32	1-11/16	2- 5/32	1-11/16	2- 7/32	1- 9/16	2-13/64	1- 9/16	2- 7/32	1- 7/16	2- 7/32	1-3/8	2- 9/32
22	1-7/8	2- 9/32	1-13/16	2- 9/32	1-13/16	2-11/32	1-11/16	2-21/64	1-11/16	2-11/32	1- 9/16	2-11/32	1-1/2	2-13/32
23	2	2-13/32	1-15/16	2-13/32	1-15/16	2-15/32	1-13/16	2-29/64	1-13/16	2-15/32	1-11/16	2-15/32	1-5/8	2-17/32
24	2-1/8	2-17/32	2- 1/16	2-17/32	2- 1/16	2-19/32	1-15/16	2-37/64	1-15/16	2-19/32	1-13/16	2-19/32	1-3/4	2-21/32
25	2-1/4	2-21/32	2- 3/16	2-21/32	2- 3/16	2-23/32	2- 1/16	2-45/64	2- 1/16	2-23/32	1-15/16	2-23/32	1-7/8	2-25/32
26	2-3/8	2-25/32	2- 5/16	2-25/32	2- 5/16	2-27/32	2- 3/16	2-53/64	2- 3/16	2-27/32	2- 1/16	2-27/32	2	2-29/32
27	2-1/2	2-29/32	2- 7/16	2-29/32	2- 7/16	2-31/32	2- 5/16	2-61/64	2- 5/16	2-31/32	2- 3/16	2-31/32	2-1/8	3- 1/32
30	2-5/8	3- 1/32	2- 9/16	3- 1/32	2- 9/16	3- 3/32	2- 7/16	3- 5/64	2- 7/16	3- 3/32	2- 5/16	3- 3/32	2-1/4	3- 5/32
31	2-3/4	3- 5/32	2-11/16	3- 5/32	2-11/16	3- 7/32	2- 9/16	3-13/64	2- 9/16	3- 7/32	2- 7/16	3- 7/32	2-3/8	3- 9/32
32	2-7/8	3- 9/32	2-13/16	3- 9/32	2-13/16	3-11/32	2-11/16	3-21/64	2-11/16	3-11/32	2- 9/16	3-11/32	2-1/2	3-13/32
33	3	3-13/32	2-15/16	3-13/32	2-15/16	3-15/32	2-13/16	3-29/64	2-13/16	3-15/32	2-11/16	3-15/32	2-5/8	3-17/32
34	3-1/8	3-17/32	3- 1/16	3-17/32	3- 1/16	3-19/32	2-15/16	3-37/64	2-15/16	3-19/32	2-13/16	3-19/32	2-3/4	3-21/32
35	3-1/4	3-21/32	3- 3/16	3-21/32	3- 3/16	3-23/32	3- 1/16	3-45/64	3- 1/16	3-23/32	2-15/16	3-23/32	2-7/8	3-25/32
36	3-3/8	3-25/32	3- 5/16	3-25/32	3- 5/16	3-27/32	3- 3/16	3-53/64	3- 3/16	3-27/32	3- 1/16	3-27/32	3	3-29/32
37	3-1/2	3-29/32	3- 7/16	3-29/32	3- 7/16	3-31/32	3- 5/16	3-61/64	3- 5/16	3-31/32	3- 3/16	3-31/32	3-1/8	4- 1/32
40	3-5/8	4- 1/32	3- 9/16	4- 1/32	3- 9/16	4- 3/32	3- 7/16	4- 5/64	3- 7/16	4- 3/32	3- 5/16	4- 3/32	3-1/4	4- 5/32
41	3-3/4	4- 5/32	3-11/16	4- 5/32	3-11/16	4- 7/32	3- 9/16	4-13/64	3- 9/16	4- 7/32	3- 7/16	4- 7/32	3-3/8	4- 9/32
42	3-7/8	4- 9/32	3-13/16	4- 9/32	3-13/16	4-11/32	3-11/16	4-21/64	3-11/16	4-11/32	3- 9/16	4-11/32	3-1/2	4-13/32
43	4	4-13/32	3-15/16	4-13/32	3-15/16	4-15/32	3-13/16	4-29/64	3-13/16	4-15/32	3-11/16	4-15/32	3-5/8	4-17/32
44	4-1/8	4-17/32	4- 1/16	4-17/32	4- 1/16	4-19/32	3-15/16	4-37/64	3-15/16	4-19/32	3-13/16	4-19/32	3-3/4	4-21/32
45	4-1/4	4-21/32	4- 3/16	4-21/32	4- 3/16	4-23/32	4- 1/16	4-45/64	4- 1/16	4-23/32	3-15/16	4-23/32	3-7/8	4-25/32
46	4-3/8	4-25/32	4- 5/16	4-25/32	4- 5/16	4-27/32	4- 3/16	4-53/64	4- 3/16	4-27/32	4- 1/16	4-27/32	4	4-29/32
47	4-1/2	4-29/32	4- 7/16	4-29/32	4- 7/16	4-31/32	4- 5/16	4-61/64	4- 5/16	4-31/32	4- 3/16	4-31/32	4-1/8	5- 1/32
50	4-5/8	5- 1/32	4- 9/16	5- 1/32	4- 9/16	5- 3/32	4- 7/16	5- 5/64	4- 7/16	5- 3/32	4- 5/16	5- 3/32	4-1/4	5- 5/32
51	4-3/4	5- 5/32	4-11/16	5- 5/32	4-11/16	5- 7/32	4- 9/16	5-13/64	4- 9/16	5- 7/32	4- 7/16	5- 7/32	4-3/8	5- 9/32
52	4-7/8	5- 9/32	4-13/16	5- 9/32	4-13/16	5-11/32	4-11/16	5-21/64	4-11/16	5-11/32	4- 9/16	5-11/32	4-1/2	5-13/32
53	5	5-13/32	4-15/16	5-13/32	4-15/16	5-15/32	4-13/16	5-29/64	4-13/16	5-15/32	4-11/16	5-15/32	4-5/8	5-17/32
54	5-1/8	5-17/32	5- 1/16	5-17/32	5- 1/16	5-19/32	4-15/16	5-37/64	4-15/16	5-19/32	4-13/16	5-19/32	4-3/4	5-21/32
55	5-1/4	5-21/32	5- 3/16	5-21/32	5- 3/16	5-23/32	5- 1/16	5-45/64	5- 1/16	5-23/32	4-15/16	5-23/32	4-7/8	5-25/32
56	5-3/8	5-25/32	5- 5/16	5-25/32	5- 5/16	5-27/32	5- 3/16	5-53/64	5- 3/16	5-27/32	5- 1/16	5-27/32	5	5-29/32
57	5-1/2	5-29/32	5- 7/16	5-29/32	5- 7/16	5-31/32	5- 5/16	5-61/64	5- 5/16	5-31/32	5- 3/16	5-31/32	5-1/8	6- 1/32
60	5-5/8	6- 1/32	5- 9/16	6- 1/32	5- 9/16	6- 3/32	5- 7/16	6- 5/64	5- 7/16	6- 3/32	5- 5/16	6- 3/32	5-1/4	6- 5/32
61	5-3/4	6- 5/32	5-11/16	6- 5/32	5-11/16	6- 7/32	5- 9/16	6-13/64	5- 9/16	6- 7/32	5- 7/16	6- 7/32	5-3/8	6- 9/32
62	5-7/8	6- 9/32	5-13/16	6- 9/32	5-13/16	6-11/32	5-11/16	6-21/64	5-11/16	6-11/32	5- 9/16	6-11/32	5-1/2	6-13/32
63	6	6-13/32	5-15/16	6-13/32	5-15/16	6-15/32	5-13/16	6-29/64	5-13/16	6-15/32	5-11/16	6-15/32	5-5/8	6-17/32
64	6-1/8	6-17/32	6- 1/16	6-17/32	6- 1/16	6-19/32	5-15/16	6-37/64	5-15/16	6-19/32	5-13/16	6-19/32	5-3/4	6-21/32
65	6-1/4	6-21/32	6- 3/16	6-21/32	6- 3/16	6-23/32	6- 1/16	6-45/64	6- 1/16	6-23/32	5-15/16	6-23/32	5-7/8	6-25/32
66	6-3/8	6-25/32	6- 5/16	6-25/32	6- 5/16	6-27/32	6- 3/16	6-53/64	6- 3/16	6-27/32	6- 1/16	6-27/32	6	6-29/32
67	6-1/2	6-29/32	6- 7/16	6-29/32	6- 7/16	6-31/32	6- 5/16	6-61/64	6- 5/16	6-31/32	6- 3/16	6-31/32	6-1/8	7- 1/32
70	6-5/8	7- 1/32	6- 9/16	7- 1/32	6- 9/16	7- 3/32	6- 7/16	7- 5/64	6- 7/16	7- 3/32	6- 5/16	7- 3/32	6-1/4	7- 5/32
71	6-3/4	7- 5/32	6-11/16	7- 5/32	6-11/16	7- 7/32	6- 9/16	7-13/64	6- 9/16	7- 7/32	6- 7/16	7- 7/32	6-3/8	7- 9/32
72	6-7/8	7- 9/32	6-13/16	7- 9/32	6-13/16	7-11/32	6-11/16	7-21/64	6-11/16	7-11/32	6- 9/16	7-11/32	6-1/2	7-13/32
73	7	7-13/32	6-15/16	7-13/32	6-15/16	7-15/32	6-13/16	7-29/64	6-13/16	7-15/32	6-11/16	7-15/32	6-5/8	7-17/32
74	7-1/8	7-17/32	7- 1/16	7-17/32	7- 1/16	7-19/32	6-15/16	7-37/64	6-15/16	7-19/32	6-13/16	7-19/32	6-3/4	7-21/32
75	7-1/4	7-21/32	7- 3/16	7-21/32	7- 3/16	7-23/32	7- 1/16	7-45/64	7- 1/16	7-23/32	6-15/16	7-23/32	6-7/8	7-25/32
76	7-3/8	7-25/32	7- 5/16	7-25/32	7- 5/16	7-27/32	7- 3/16	7-53/64	7- 3/16	7-27/32	7- 1/16	7-27/32	7	7-29/32
77	7-1/2	7-29/32	7- 7/16	7-29/32	7- 7/16	7-31/32	7- 5/16	7-61/64	7- 5/16	7-31/32	7- 3/16	7-31/32	7-1/8	8- 1/32
80	7-5/8	8- 1/32	7- 9/16	8- 1/32	7- 9/16	8- 3/32	7- 7/16	8- 5/64	7- 7/16	8- 3/32	7- 5/16	8- 3/32	7-1/4	8- 5/32
81					7-11/16	8- 7/32					7- 7/16	8- 7/32		
82											7- 9/16	8-11/32		
83											7-11/16	8-15/32		
84											7-13/16	8-19/32		
85							8- 1/16	8-45/64			7-15/16	8-23/32		
86											8- 1/16	8-27/32		
87											8- 3/16	8-31/32		

APPROVED 29 Mar 44 REVISED ① 24 Mar 52

PROCUREMENT SPECIFICATION

① MIL-B-6812

AIR FORCE-NAVY AERONAUTICAL STANDARD

① BOLT - MACHINE, CLOSE TOLERANCE, AIRCRAFT

AN173 THRU AN186

①

SHEET 2 OF 4

[illegible]

DASH NO.	AN180		AN182		AN184		AN186	
	GRIP ±1/64	LENGTH +1/32 -1/64	GRIP ±1/64	LENGTH +1/32 -1/64	GRIP ±1/64	LENGTH +1/32 -1/64	GRIP ±1/64	LENGTH +1/32 -1/64
7	1/16	1- 1/64						
10	3/16	1- 9/64	1/16	1- 5/32				
11	5/16	1-17/64	3/16	1- 9/32	1/16	1- 5/16		
12	7/16	1-25/64	5/16	1-13/32	3/16	1- 7/16	1/8	1-1/2
13	9/16	1-33/64	7/16	1-17/32	5/16	1- 9/16	1/4	1-5/8
14	11/16	1-41/64	9/16	1-21/32	7/16	1-11/16	3/8	1-3/4
15	13/16	1-49/64	11/16	1-25/32	9/16	1-13/16	1/2	1-7/8
16	15/16	1-57/64	13/16	1-29/32	11/16	1-15/16	5/8	2
17	1- 1/16	2- 1/64	15/16	2- 1/32	13/16	2- 1/16	3/4	2-1/8
20	1- 3/16	2- 9/64	1- 1/16	2- 5/32	15/16	2- 3/16	7/8	2-1/4
21	1- 5/16	2-17/64	1- 3/16	2- 9/32	1- 1/16	2- 5/16	1	2-3/8
22	1- 7/16	2-25/64	1- 5/16	2-13/32	1- 3/16	2- 7/16	1-1/8	2-1/2
23	1- 9/16	2-33/64	1- 7/16	2-17/32	1- 5/16	2- 9/16	1-1/4	2-5/8
24	1-11/16	2-41/64	1- 9/16	2-21/32	1- 7/16	2-11/16	1-3/8	2-3/4
25	1-13/16	2-49/64	1-11/16	2-25/32	1- 9/16	2-13/16	1-1/2	2-7/8
26	1-15/16	2-57/64	1-13/16	2-29/32	1-11/16	2-15/16	1-5/8	3
27	2- 1/16	3- 1/64	1-15/16	3- 1/32	1-13/16	3- 1/16	1-3/4	3-1/8
30	2- 3/16	3- 9/64	2- 1/16	3- 5/32	1-15/16	3- 3/16	1-7/8	3-1/4
31	2- 5/16	3-17/64	2- 3/16	3- 9/32	2- 1/16	3- 5/16	2	3-3/8
32	2- 7/16	3-25/64	2- 5/16	3-13/32	2- 3/16	3- 7/16	2-1/8	3-1/2
33	2- 9/16	3-33/64	2- 7/16	3-17/32	2- 5/16	3- 9/16	2-1/4	3-5/8
34	2-11/16	3-41/64	2- 9/16	3-21/32	2- 7/16	3-11/16	2-3/8	3-3/4
35	2-13/16	3-49/64	2-11/16	3-25/32	2- 9/16	3-13/16	2-1/2	3-7/8
36	2-15/16	3-57/64	2-13/16	3-29/32	2-11/16	3-15/16	2-5/8	4
37	3- 1/16	4- 1/64	2-15/16	4- 1/32	2-13/16	4- 1/16	2-3/4	4-1/8
40	3- 3/16	4- 9/64	3- 1/16	4- 5/32	2-15/16	4- 3/16	2-7/8	4-1/4
41	3- 5/16	4-17/64	3- 3/16	4- 9/32	3- 1/16	4- 5/16	3	4-3/8
42	3- 7/16	4-25/64	3- 5/16	4-13/32	3- 3/16	4- 7/16	3-1/8	4-1/2
43	3- 9/16	4-33/64	3- 7/16	4-17/32	3- 5/16	4- 9/16	3-1/4	4-5/8
44	3-11/16	4-41/64	3- 9/16	4-21/32	3- 7/16	4-11/16	3-3/8	4-3/4
45	3-13/16	4-49/64	3-11/16	4-25/32	3- 9/16	4-13/16	3-1/2	4-7/8
46	3-15/16	4-57/64	3-13/16	4-29/32	3-11/16	4-15/16	3-5/8	5
47	4- 1/16	5- 1/64	3-15/16	5- 1/32	3-13/16	5- 1/16	3-3/4	5-1/8
50	4- 3/16	5- 9/64	4- 1/16	5- 5/32	3-15/16	5- 3/16	3-7/8	5-1/4
51	4- 5/16	5-17/64	4- 3/16	5- 9/32	4- 1/16	5- 5/16	4	5-3/8
52	4- 7/16	5-25/64	4- 5/16	5-13/32	4- 3/16	5- 7/16	4-1/8	5-1/2
53	4- 9/16	5-33/64	4- 7/16	5-17/32	4- 5/16	5- 9/16	4-1/4	5-5/8
54	4-11/16	5-41/64	4- 9/16	5-21/32	4- 7/16	5-11/16	4-3/8	5-3/4
55	4-13/16	5-49/64	4-11/16	5-25/32	4- 9/16	5-13/16	4-1/2	5-7/8
56	4-15/16	5-57/64	4-13/16	5-29/32	4-11/16	5-15/16	4-5/8	6
57	5- 1/16	6- 1/64	4-15/16	6- 1/32	4-13/16	6- 1/16	4-3/4	6-1/8
60	5- 3/16	6- 9/64	5- 1/16	6- 5/32	4-15/16	6- 3/16	4-7/8	6-1/4
61	5- 5/16	6-17/64	5- 3/16	6- 9/32	5- 1/16	6- 5/16	5	6-3/8
62	5- 7/16	6-25/64	5- 5/16	6-13/32	5- 3/16	6- 7/16	5-1/8	6-1/2
63	5- 9/16	6-33/64	5- 7/16	6-17/32	5- 5/16	6- 9/16	5-1/4	6-5/8
64	5-11/16	6-41/64	5- 9/16	6-21/32	5- 7/16	6-11/16	5-3/8	6-3/4
65	5-13/16	6-49/64	5-11/16	6-25/32	5- 9/16	6-13/16	5-1/2	6-7/8
66	5-15/16	6-57/64	5-13/16	6-29/32	5-11/16	6-15/16	5-5/8	7
67	6- 1/16	7- 1/64	5-15/16	7- 1/32	5-13/16	7- 1/16	5-3/4	7-1/8
70	6- 3/16	7- 9/64	6- 1/16	7- 5/32	5-15/16	7- 3/16	5-7/8	7-1/4
71	6- 5/16	7-17/64	6- 3/16	7- 9/32	6- 1/16	7- 5/16	6	7-3/8
72	6- 7/16	7-25/64	6- 5/16	7-13/32	6- 3/16	7- 7/16	6-1/8	7-1/2
73	6- 9/16	7-33/64	6- 7/16	7-17/32	6- 5/16	7- 9/16	6-1/4	7-5/8
74	6-11/16	7-41/64	6- 9/16	7-21/32	6- 7/16	7-11/16	6-3/8	7-3/4
75	6-13/16	7-49/64	6-11/16	7-25/32	6- 9/16	7-13/16	6-1/2	7-7/8
76	6-15/16	7-57/64	6-13/16	7-29/32	6-11/16	7-15/16	6-5/8	8
77	7- 1/16	8- 1/64	6-15/16	8- 1/32	6-13/16	8- 1/16	6-3/4	8-1/8
80	7- 3/16	8- 9/64	7- 1/16	8- 5/32	6-15/16	8- 3/16	6-7/8	8-1/4
81							7	8-3/8
82							7-1/8	8-1/2

① (a) MARKING MAY BE EITHER RAISED OR DEPRESSED.
THE HEIGHT OR DEPTH OF THE MARKINGS SHALL NOT EXCEED THE FOLLOWING VALUES:
.025 INCH FOR 9/16 DIAMETER BOLTS AND SMALLER;
.030 INCH FOR 5/8 DIAMETER BOLTS AND LARGER.
HEAD MARKING MAY INCLUDE MANUFACTURER'S MARKINGS.

① (b) GRIP LENGTH OF BOLTS SHALL BE MEASURED FROM THE UNDERSIDE OF THE HEAD TO THE END OF THE FULL CYLINDRICAL PORTION OF THE SHANK. COMPLETE THREADS SHALL BEGIN WITHIN TWO THREAD PITCH MAXIMUM. TWO THREAD PITCH MAXIMUM MAY CONSIST OF INCOMPLETE THREAD OR EXTRUSION ANGLE.

① (c) CORROSION RESISTING STEEL BOLTS AND ALUMINUM ALLOY BOLTS INACTIVE FOR DESIGN AFTER 24 MARCH 1952.
① FOR DEFINITION AND APPLICATION OF DRAWING STATUS NOTES, SEE ANA BULLETIN NO. 337.

APPROVED 29 Mar 49 REVISED ① 24 Mar 52

PROCUREMENT SPECIFICATION
① MIL-B-6812

AIR FORCE-NAVY AERONAUTICAL STANDARD

① BOLT-MACHINE, CLOSE TOLERANCE, AIRCRAFT

AN173 THRU AN186

① SHEET 3 OF 4

NOTICE.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

NOTE.—This drawing was approved by joint action of the Air Force and Navy Department as the Air Force-Navy standard for this product. This drawing supersedes all antecedent standard drawings for the same product and shall become effective for the procurement of aeronautical supplies, or for use in new design, not later than six months after the latest date of approved sheets.

BASIC AN PART NO.	THREAD T	RATED STRENGTH (POUNDS)					
		ULTIMATE TENSILE AT ROOT DIA		YIELD TENSILE AT ROOT DIA		SINGLE SHEAR AT FULL DIA	
		STEEL	AL ALLOY	STEEL	AL ALLOY	STEEL	AL ALLOY
AN173	NO. 10-32 NF-3A	2 210	1 100	1 690	710	2 125	990
AN174	1/4 -28 UNF-3A	4 090	2 030	3 130	1 310	3 680	1 715
AN175	5/16-24 UNF-3A	6 500	3 220	4 980	2 080	5 750	2 685
AN176	3/8 -24 UNF-3A	10 100	5 020	7 740	3 240	8 280	3 870
AN177	7/16-20 UNF-3A	13 600	6 750	10 430	4 350	11 250	5 250
AN178	1/2 -20 UNF-3A	18 500	9 180	14 170	5 920	14 700	6 850
AN179	9/16-18 UNF-3A	23 600	11 700	18 100	7 550	18 700	8 700
AN180	5/8 -18 UNF-3A	30 100	14 900	23 080	9 610	23 000	10 750
AN182	3/4 -16 UNF-3A	44 000	21 800	33 730	14 100	33 150	15 500
AN184	7/8 -14 UNF-3A	60 000	29 800	46 000	19 200	45 050	21 050
AN186	1 -14 NF-3A	80 700	40 000	61 870	25 800	58 900	27 500

APPROVED 24 Mar 52

REVISED

PROCUREMENT SPECIFICATION

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